

1. Administrative & Study Identification

Full Study Title: A Phase II, randomized, open-label, multi-center study of JSB462 (luxdegalutamide) in combination with lutetium (177Lu) vipivotide tetraxetan in adult male patients with PSMA-positive metastatic castration resistant prostate cancer (mCRPC) **Short Title/Acronym:** LuxAR-02 **Protocol Number:** CJSB462B12201 **NCT Number:** NCT07047118 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** JSB462 (luxdegalutamide) and Lutetium (177Lu) vipivotide tetraxetan (AAA617 / Lu-PSMA-617) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (PSA90 rate) and safety of the combination of JSB462 (luxdegalutamide) + Lu-PSMA-617 compared with Lu-PSMA-617 alone, and to select the recommended dose of the combination for Phase III. **Secondary Objectives:** To evaluate radiographic progression-free survival (rPFS), overall survival (OS), pharmacokinetic (PK) data, and objective response rates (ORR).

2.2 Background & Rationale Treatment resistance in mCRPC is frequently driven by androgen receptor (AR) signaling reactivation, creating an unmet need for new therapies targeting the AR pathway. Luxdegalutamide is an oral AR degrader (PROTAC) that has shown a manageable safety profile and encouraging clinical activity in heavily pretreated mCRPC patients. Lu-PSMA-617 is an approved PSMA-targeted radioligand therapy. This study evaluates whether combining these two agents improves treatment efficacy based on their potentially complementary mechanisms of action.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Lu-PSMA-617 alone) **Blinding:** Open-label **Randomization:** Randomized (5:5:3 ratio across three arms)

3.2 Planned Enrollment & Duration Target \$N\$: 130 adult male patients **Number of Sites:** Global, multi-center **Estimated Study Start:** 03-Jul-2025 **Estimated Primary Completion:** 07-Jan-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult male participants (≥ 18 years) with histologically and/or cytologically confirmed adenocarcinoma of the prostate. 2. Participants must be [68Ga]Ga-PSMA-11 PET/CT scan positive and eligible as determined by the sponsor's central reader. 3. Prior exposure to at least one second-generation Androgen Receptor Pathway Inhibitor (ARPI) in the metastatic/advanced setting. 4. At least 1 bone or visceral metastatic lesion present on baseline imaging obtained ≤ 28 days prior to study treatment. 5. Eastern Cooperative Oncology Group (ECOG) Performance Status ≤ 2 .

4.2 Exclusion Criteria 1. Participants with mixed histology (neuroendocrine) prostate cancer. 2. Previous treatment with more than 2 taxane regimens. 3. Inability to swallow oral medication.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm 1:** Luxdegalutamide 100 mg QD + Lu-PSMA-617 (7.4 GBq Q6W for up to 6 doses) * **Arm 2:** Luxdegalutamide 300 mg QD + Lu-PSMA-617 (7.4 GBq Q6W for up to 6 doses) * **Arm 3 (Control):** Lu-PSMA-617 alone (7.4 GBq Q6W for up to 6 doses)

Route of Administration: Oral (JSB462) and Intravenous (Lu-PSMA-617) **Duration of Treatment:** JSB462 is taken daily and continuously until disease progression or unacceptable toxicity. Lu-PSMA-617 is administered every 6 weeks for up to 6 doses.

5.2 Concomitant & Prohibited Therapy Permitted: Prior PARPi and/or immune checkpoint inhibitors are allowed if the patient has previous exposure. Standard of care treatments for co-morbidities. **Prohibited:** Concurrent anti-cancer systemic therapies outside of the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Imaging (CT/MRI/PET) Baseline
Day 1	Randomization, First Dose JSB462	(14-day lead-in for Arms 1 & 2) Interim	Q6W Lu-PSMA-617 administration, Safety Labs, Efficacy Assessment (PSA) End of Study
			Variable Disease Progression, Exit Interview, 30-Day Safety Follow-up, Long-Term Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy: Prostate Specific Antigen 90 (PSA90) Rate from Baseline at any point, confirmed by a 2nd PSA $\geq$ 3 weeks without progression in between. **Safety:** Incidence rate of adverse events (AEs). **Measurement Method:** Serum PSA laboratory analysis and clinical safety assessments.

7.2 Secondary & Exploratory Endpoints * Radiographic progression-free survival (rPFS) per PCWG3-modified RECIST v1.1 * Objective Response Rate (ORR) * Overall Survival (OS) * Pharmacokinetics (PK) of the combination regimen

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical assessments, patient reporting, and laboratory evaluations during visits. **Serious Adverse Events (SAEs):** Standard reporting timelines per institutional and sponsor guidelines. **Data Monitoring Committee (DMC):** External/Internal committee established by Novartis to oversee trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy measures; Safety Set for safety endpoints. **Sample Size Justification:** $N = 130$ randomized in a 5:5:3 ratio, powered to select the appropriate dose combination for Phase III. **Handling of Missing Data:** Standard protocol-defined rules (e.g., censoring for survival endpoints, missing assessments treated as non-responders where applicable).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized and onsite monitoring of clinical data and safety metrics. **Audit Trail:** eCRF data entry, central reader review for [68Ga]Ga-PSMA-11 PET/CT scans, and strict radiation handling protocols.

11. Study Identifiers & Governance

Primary ID: CJSB462B12201 **Registry Number:** NCT07047118 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** LuxAR-02 **Official Regulatory Title:** A Phase II, randomized, open-label, multi-center study of JSB462 (luxdegalutamide) in combination with lutetium (^{177}Lu) vipivotide tetraxetan in adult male

patients with PSMA-positive metastatic castration resistant prostate cancer (mCRPC)

12. Clinical Framework (Trial Specific)

Note: Adapted from template for mCRPC Primary Condition:
Metastatic Castration Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** PSMA-positive on [68Ga]Ga-PSMA-11 PET/CT scan **Medical Methodology:** Androgen Receptor Pathway Degradar + Radioligand Therapy **Optimization Target:** Prostate Specific Antigen 90 (PSA90) response

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced mCRPC **Education Protocol (HCP):** Radiation safety protocol and ARPI management **Education Protocol (Patient):** Radiation isolation/safety guidelines and oral dosing compliance **Quality Audit Mechanism:** Central imaging review (Sponsor's central reader) and continuous AE tracking

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression and long-term overall survival follow-up **Onsite Visit Cadence:** Every 6 weeks (Q6W) during the IV radioligand treatment phase **Remote Monitoring Frequency:** Between treatment cycles as per institutional guidelines **Checklist Review Interval:** Post-treatment safety follow-up at 30 days, followed by long-term follow-up

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				